2023

Hootsuite Accelerates Innovation While Improving Resiliency and Efficiency on AWS

Learn how social media management provider Hootsuite successfully consolidated 2,000 Amazon EC2 instances using Amazon EKS.

Overview | Opportunity | Solution | Outcome | AWS Services Used

2,000+ to <500 reduction

in Amazon EC2 instances

28% reduction

in security incidents year over year

Overview

[Hootsuite](), a provider of social media management products and services, empowers brands and organizations around the world to harness the power of social. The company supports its tools using a robust set of microservices for optimal agility and scalability. As time passed, however, Hootsuite's sprawling infrastructure became difficult to manage.

In its quest for streamlined operations, Hootsuite turned to Amazon Web Services (AWS) and embraced a containerized approach. This shift helped Hootsuite optimize its AWS footprint, which reduced infrastructure costs, improved resiliency, and accelerated development productivity.

Opportunity | Using Amazon EKS to Reduce Operational Overhead and Modernize Hootsuite's Technology Landscape

Founded in 2008, Hootsuite offers a suite of tools to help customers unlock the value of their social relationships, from scheduling posts to running ads and tracking performance metrics. As an early adopter of microservices architecture, the company supported its production infrastructure using over 2,000 Amazon Elastic Compute Cloud (Amazon EC2) instances, which offer secure and resizable compute capacity for virtually any workload. However, efficiently managing this extensive Amazon EC2 estate meant acquiring new tools.

"A microservices architecture presents many benefits but also creates new challenges," says L ector of platform and infrastructure at Hootsuite. "As the number of services grows, the complexity omes harder to manage those individual services, and dependency is often not clearly defined. When we run into a problem in production, it can be difficult to troubleshoot."

After the Cloud Native Computing Foundation endorsed Kubernetes, an open-source container orchestration system, in 2016, Hootsuite began to explore containerization technology as a potential solution. Following an initial proof of concept, the company decided to adopt a self-managed Kubernetes environment on AWS.

In 2018, Amazon Elastic Kubernetes Service (Amazon EKS), a managed Kubernetes service in the AWS cloud and on-premises data centers, was launched. Recognizing the potential of this service, Hootsuite gradually migrated its entire production environment to Amazon EKS.

"To improve business agility and development productivity, you need a consistent environment to operate in," says Guo. "Without that consistency, people are going to spend more time trying to figure things out, which is going to incur overhead costs. With Amazon EKS, we have been able to build an enterprise-scale managed Kubernetes platform to support our business and drive innovation."

Solution | Achieving a Cost Savings of 40% While Improving Resiliency and Latency

By the beginning of 2022, Hootsuite successfully migrated virtually 100 percent of its production services to Amazon EKS. Today, this environment runs over 700 different services. This nearly 7-year project involved a deep collaboration between Hootsuite and the Amazon EKS team, which helped the company optimize the new environment and consolidate its number of Amazon EC2 instances.

"Using Amazon EKS, we were able to reduce our Amazon EC2 footprint from over 2,000 instances to less than 500 in a few years," says Guo. "From a cost savings perspective, our Amazon EC2 expenses this year are about 40 percent less than before."

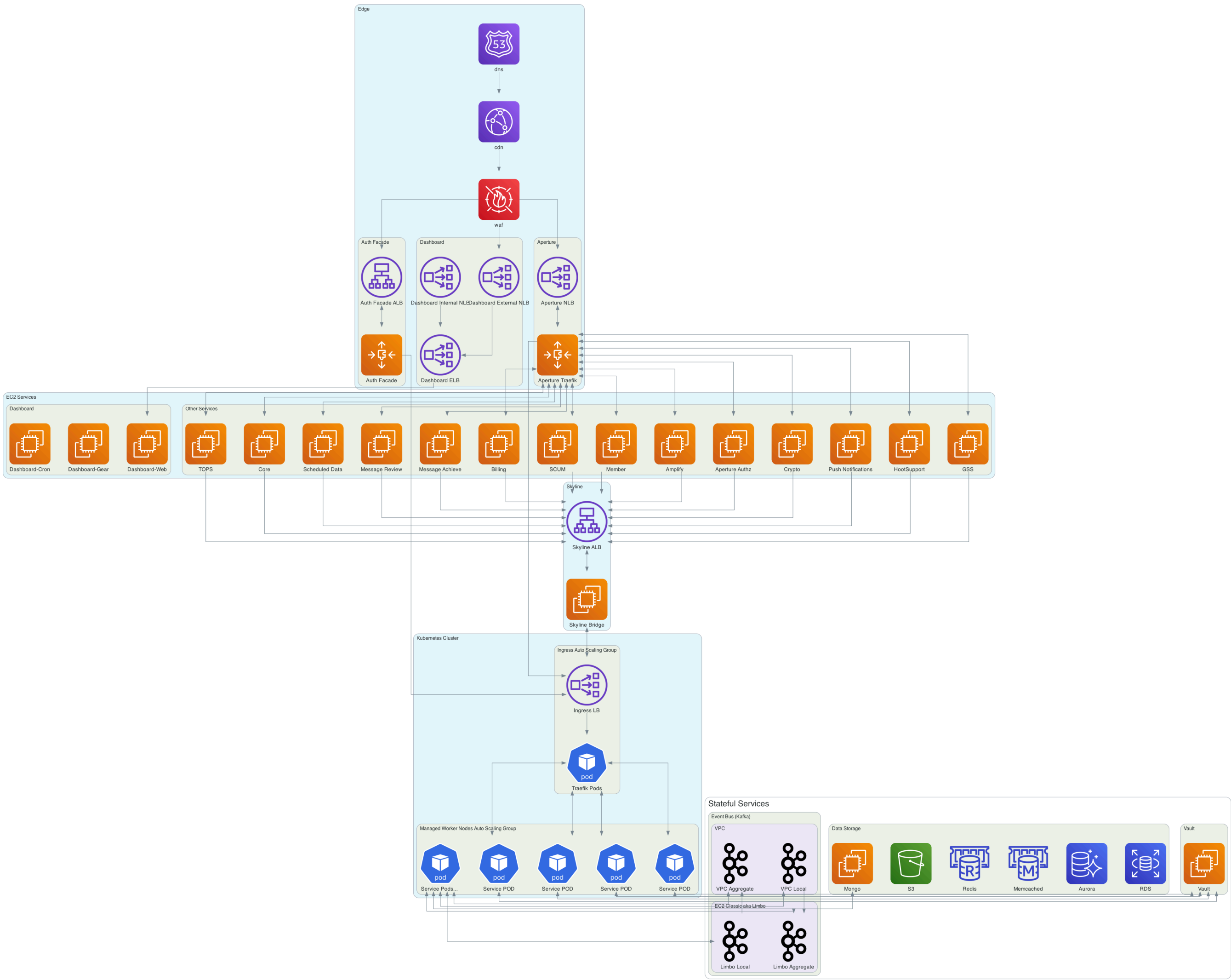
Reliability has been a cornerstone benefit of transitioning to Amazon EKS. In the past, outages would impact both Hootsuite and its customers. System downtime would prevent clients from logging in, publish … media posts, or even sending out scheduled messages. Hootsuite sorts its most serious incidents into … SEV1, which entails a complete system outage that causes Hootsuite to be down for all custo… which impacts major product functionality and prevents a significant portion of customers from performing core tasks. By migrating to Amazon EKS, Hootsuite has reduced the number of SEV1 and SEV2 incidents by 57 percent.

By streamlining its architecture on AWS, Hootsuite has also bolstered its security posture. The consolidated environment has helped the company minimize its exposure to potential threats. With a smaller footprint, Hootsuite can enforce policies and protocols more uniformly, which has facilitated stronger governance.

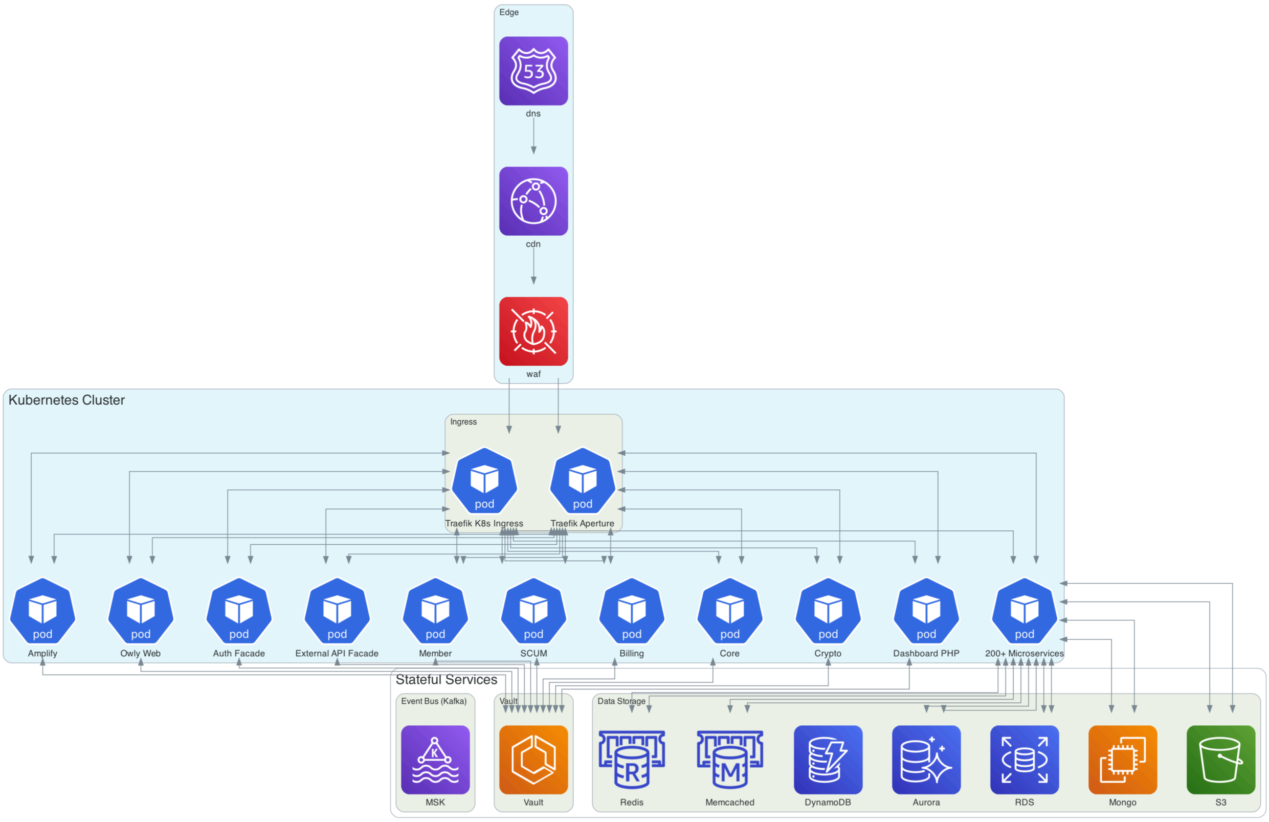
“One of the reasons that we wanted to minimize our AWS footprint was to reduce the attack surface, along with gaining more consistency and standardization,” says Guo. “Our total number of overall incidents this year is about 28 percent less than last year.”

By simplifying its AWS estate, Hootsuite has unlocked key efficiencies for its development teams. “With Kubernetes running on Amazon EKS, we can bring all of our microservices into one environment, and operate as one,” says Guo. “This is a huge benefit to all of our teams and service owners. We have a much deeper shared understanding of how our service is actually operating.” With less time spent on understanding the intricacies of the environment, Hootsuite teams have more opportunities to explore new projects and optimize their services to unlock additional efficiencies, including time and cost savings.

Architecture Diagram



Before



After

Outcome | Adopting New Technologies to Deliver Product Innovation with Security and Resiliency at the Core

By embracing a containerized approach, Hootsuite has not only streamlined its expansive AWS estate but also achieved significant cost savings and boosted system resiliency. Looking ahead, it plans to fine-tune its Amazon EKS environment with new technologies. The company is in the process of deploying Bottlerocket, a Linux-based open-source operating system developed by AWS, across its entire Amazon EC2 fleet. Hootsuite will also expand its usage of Karpenter, an open-source scaling and node provisioning system.

"At Hootsuite, technology is always a core part of our company," says Guo. "We often encourage teams to find the best solution for the problem at hand. We've definitely conducted a lot of trial and error over the years, but we're in a much better place today on AWS."

About Hootsuite

Hootsuite helps customers harness the power of social media to ignite their brands and businesses. Its expertise empowers thousands of organizations to strategically grow their brands, businesses, and customer relationships.

AWS Services Used

Amazon Elastic Kubernetes Service

Amazon Elastic Kubernetes Service (Amazon EKS) is a managed Kubernetes service to run Kubernetes in the AWS cloud and on-premises data centers.

Learn more »

Amazon Elastic Compute Cloud

Amazon Elastic Compute Cloud (Amazon EC2) offers the broadest and deepest compute platform, with over 750 instances and choice of the latest processor, storage, networking, operating system, and purchase model to help you best match the needs of your workload.

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